

University of Texas at Austin, Jackson School of Geosciences
GEO 325C/398C, 2026
Continuum Mechanics

Midterm Exam – ANSWERS [25 pts]

- (1) **The One with Index Notation [8 pts]:** The shear stress, $\mathbf{t}^\perp$ on a plane with normal, $\mathbf{n}$, can be computed in three ways:

- $\mathbf{t}^\perp = \mathbf{t} - (\mathbf{t} \cdot \mathbf{n}) \mathbf{n}$
- $\mathbf{t}^\perp = \mathbf{P}^\perp \mathbf{t} = (\mathbf{I} - \mathbf{n} \otimes \mathbf{n}) \mathbf{t}$
- $\mathbf{t}^\perp = -(\mathbf{t} \times \mathbf{n}) \times \mathbf{n}$

Use index notation to show that $(\mathbf{I} - \mathbf{n} \otimes \mathbf{n}) \mathbf{t} = -(\mathbf{t} \times \mathbf{n}) \times \mathbf{n}$.

Use vectors with the following indices: $\mathbf{t} = t_i \mathbf{e}_i$, $\mathbf{n} = n_j \mathbf{e}_j$ and $\mathbf{n} = n_k \mathbf{e}_k$. The identity is given by $\mathbf{I} = \delta_{jk} \mathbf{e}_j \otimes \mathbf{e}_k$ the dyadic product $\mathbf{n} \otimes \mathbf{n} = n_j n_k \mathbf{e}_j \otimes \mathbf{e}_k$. The frame identities are $\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$ and $\mathbf{e}_i \times \mathbf{e}_j = \epsilon_{ijk} \mathbf{e}_k$. The $\delta\epsilon$ -identity is given by $\epsilon_{ijk} \epsilon_{pqk} = \delta_{ip} \delta_{jq} - \delta_{iq} \delta_{jp}$.

The left hand side becomes

$$\begin{aligned} (\mathbf{I} - \mathbf{n} \otimes \mathbf{n}) \mathbf{t} &= (\delta_{jk} - n_j n_k) (\mathbf{e}_j \otimes \mathbf{e}_k) (t_i \mathbf{e}_i), \\ &= (\delta_{jk} - n_j n_k) t_i (\mathbf{e}_j \otimes \mathbf{e}_k) \mathbf{e}_i, \\ &= (\delta_{jk} - n_j n_k) t_i (\mathbf{e}_k \cdot \mathbf{e}_i) \mathbf{e}_j, \\ &= (\delta_{jk} - n_j n_k) t_i \delta_{ki} \mathbf{e}_j, \\ &= \delta_{jk} \delta_{ki} t_i \mathbf{e}_j - \delta_{ki} n_j n_k t_i \mathbf{e}_j, \\ &= t_k \mathbf{e}_k - n_i t_i n_j \mathbf{e}_j, \\ &= \mathbf{t} - (\mathbf{t} \cdot \mathbf{n}) \mathbf{n}. \end{aligned}$$

The right hand side becomes

$$\begin{aligned} -\mathbf{t} \times \mathbf{n} \times \mathbf{n} &= -(\epsilon_{ijk} t_i n_j \mathbf{e}_k) \times (n_p \mathbf{e}_p), \\ &= -\epsilon_{ijk} t_i n_j n_p \mathbf{e}_k \times \mathbf{e}_p, \\ &= -\epsilon_{ijk} \epsilon_{kpq} t_i n_j n_p \mathbf{e}_q, \\ &= -\epsilon_{ijk} \epsilon_{pqk} t_i n_j n_p \mathbf{e}_q, \\ &= -(\delta_{ip} \delta_{jq} - \delta_{iq} \delta_{jp}) t_i n_j n_p \mathbf{e}_q, \\ &= -\delta_{ip} \delta_{jq} t_i n_j n_p \mathbf{e}_q + \delta_{iq} \delta_{jp} t_i n_j n_p \mathbf{e}_q, \\ &= -t_i n_i n_j \mathbf{e}_j + n_j n_j t_i \mathbf{e}_i, \\ &= -(\mathbf{t} \cdot \mathbf{n}) \mathbf{n} + (\mathbf{n} \cdot \mathbf{n}) \mathbf{t}, \\ &= \mathbf{t} - (\mathbf{t} \cdot \mathbf{n}) \mathbf{n} \end{aligned}$$

- (2) **The One with Rotations [total 9 pts]:** In geoscience a fault plane is defined by its strike, ϕ , and its dip angle, θ . The strike is the angle from north and the dip the angle from the horizontal in the direction perpendicular from the strike. The normal to any fault plane can be found from strike and dip by two subsequent rotations, $\mathbf{Q}_S$ and $\mathbf{Q}_D$, of a unit vector pointing upward. In a North-East-Down (NED) reference frame, $\{\mathbf{e}_i\}$, these rotations are defined as $\mathbf{Q}_D = \mathbf{Q}(\mathbf{e}_1, \theta)$ and $\mathbf{Q}_S = \mathbf{Q}(\mathbf{e}_3, \phi)$, where

$$\mathbf{Q}(\mathbf{r}, \vartheta) = \mathbf{r} \otimes \mathbf{r} + \cos(\vartheta) (\mathbf{I} - \mathbf{r} \otimes \mathbf{r}) + \sin(\vartheta) \mathbf{R}(\mathbf{r})$$

is the Euler representation of a finite rotation tensor and axial tensor is given by

$$\mathbf{R}(\mathbf{r}) = \epsilon_{ijk} r_j \mathbf{e}_i \otimes \mathbf{e}_k = \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix}$$

For for a fault with strike $\phi = \pi/2$ (90°) and dip $\theta = \pi/2$ (90°) do the following:

(a) Make a 3D sketch of the fault and its normal [2 pts].

(b) Compute $\mathbf{Q}_S$ and $\mathbf{Q}_D$ [4 pts].

We have the following rotation matrices

$$\mathbf{Q}_S = Q(\mathbf{e}_3, \phi) = \mathbf{e}_3 \otimes \mathbf{e}_3 + \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$\mathbf{Q}_D = Q(\mathbf{e}_1, \theta) = \mathbf{e}_1 \otimes \mathbf{e}_1 + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

(c) Show that the order of rotations matters by computing both $\mathbf{n}_1 = \mathbf{Q}_S \mathbf{Q}_D(-\mathbf{e}_3)$ and $\mathbf{n}_2 = \mathbf{Q}_D \mathbf{Q}_S(-\mathbf{e}_3)$ [2 pts].

We have the following two normals

$$\mathbf{n}_1 = \mathbf{Q}_S \mathbf{Q}_D(-\mathbf{e}_3) = \begin{bmatrix} -1 & 0 & 0 \end{bmatrix}^T,$$

$$\mathbf{n}_2 = \mathbf{Q}_D \mathbf{Q}_S(-\mathbf{e}_3) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T.$$

(d) Compare the answers with your sketch and determine the correct normal and hence the correct order of rotations [1 pt].

The correct normal is $\mathbf{n}_1$.

(3) **The One with the Stress** [total 8 pts]: The representation of stress tensor in the NED frame near the SAFOD borehole is approximately

$$\boldsymbol{\sigma} = \begin{bmatrix} 110 & 5 & 0 \\ 5 & 50 & 0 \\ 0 & 0 & 50 \end{bmatrix} \text{ MPa.}$$

The representation of the normal to the San Andreas fault in the NED frame near the SAFOD borehole is approximately given by

$$\mathbf{n} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

These values have been modified to give answers with round numbers. Please do not approximate $\sqrt{2}$ as a decimal number.

(a) Compute the traction, $\mathbf{t} = \boldsymbol{\sigma} \mathbf{n}$, on the fault [2 pts].

The traction on the fault is given by

$$\mathbf{t} = \frac{1}{\sqrt{2}} \begin{bmatrix} 115 \\ 55 \\ 0 \end{bmatrix}.$$

(b) Compute the projection of the traction parallel, $\mathbf{t}^\parallel$, and perpendicular, $\mathbf{t}^\perp$, to the fault normal, $\mathbf{n}$ [4 pts].

The traction on the fault is given by

$$\mathbf{t}^\parallel = (\mathbf{n} \cdot \mathbf{t}) \mathbf{n} = \frac{1}{\sqrt{2}} \begin{bmatrix} 85 \\ 85 \\ 0 \end{bmatrix},$$

$$\mathbf{t}^\perp = \mathbf{t} - \mathbf{t}^\parallel = \frac{1}{\sqrt{2}} \begin{bmatrix} 30 \\ -30 \\ 0 \end{bmatrix}.$$

- (c) Compute the normal, $\sigma_n = |\mathbf{t}^{\parallel}|$ and shear stress, $\tau = |\mathbf{t}^{\perp}|$, on the San Andreas fault near the SAFOD borehole [**2 pts**].

The normal and shear stresses are $\sigma_n = 85$ MPa and $\tau = 30$ MPa.